

Simulation-Based Evaluation of Geometric and Particle Filters for Multi-Camera UAV Tracking

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Abstract

Multi-camera vision-based tracking systems provide a promising alternative to radar in urban environments where line-of-sight constraints degrade traditional sensing. This work presents a simulation-based evaluation of particle filters (PF), multiplicative extended Kalman filter (MEKF), invariant extended Kalman filter (InEKF), and feedback particle filter (FPF) within a unified multi-camera tracking framework. A consistent simulation environment is developed using a calibrated camera network and perspective projection model with noisy image measurements. Performance is evaluated across multiple tracking scenarios. Preliminary results indicate that PF exhibits strong robustness under ambiguity, while geometric filters offer computational efficiency and stability in moderate noise regimes. The FPF demonstrates promising behavior but remains sensitive to gain approximation.

Introduction

Urban airspace monitoring for unmanned aerial systems (UAS) presents significant challenges due to occlusions and limited line-of-sight in dense environments. Camera networks offer a scalable alternative for tracking aerial targets by combining multi-view measurements to estimate trajectories. This work evaluates multiple filtering approaches under a unified simulation framework.

Methodology

The target is modeled using a constant-velocity state with position and velocity components. Measurements are generated via perspective projection into multiple cameras with Gaussian noise. Filters evaluated include PF, MEKF, InEKF, and FPF.

Simulation Setup

A four-camera setup surrounds a tracking volume. Targets follow linear, maneuvering, and crossing trajectories. Noise and initialization uncertainty are introduced to evaluate robustness.

Preliminary Results

PF demonstrates robustness in nonlinear and ambiguous scenarios. MEKF is efficient but sensitive to initialization. InEKF improves consistency, while FPF shows potential advantages with sensitivity to gain approximation.

Future Work

Future work includes Monte Carlo evaluation, improved FPF gain approximation, and integration with real-world data.